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Class : D15A
Roll : 40

Experiment 6 :Implementation of K-Means Clustering Using Python

Aim

To implement the K-Means clustering algorithm using Python on the Seeds dataset and analyze the clustering results using the Elbow Method, Silhouette Score, Inertia, Davies-Bouldin Index, and data visualization.

Theory

Clustering:

Clustering is an unsupervised machine learning technique used to group similar data points based on their characteristics. It identifies patterns and structures in a dataset without requiring predefined labels. Common clustering algorithms include K-Means, DBSCAN, and Hierarchical Clustering. Clustering is widely used in customer segmentation, anomaly detection, image analysis, and data mining.

K-Means Clustering:

K-Means is an unsupervised learning algorithm that divides a dataset into a predefined number of clusters (K). It assigns each data point to the nearest cluster centroid and repeatedly updates the centroids until the clusters stabilize. The main objective is to minimize the variation within each cluster.

Seeds Dataset:

The Seeds dataset contains 210 instances of wheat seeds belonging to three varieties: Kama, Rosa, and Canadian. Each instance has seven numerical attributes: area, perimeter, compactness, length of kernel, width of kernel, asymmetry coefficient, and length of kernel groove. These attributes help identify similarities among seeds through clustering.

Evaluation Metrics:

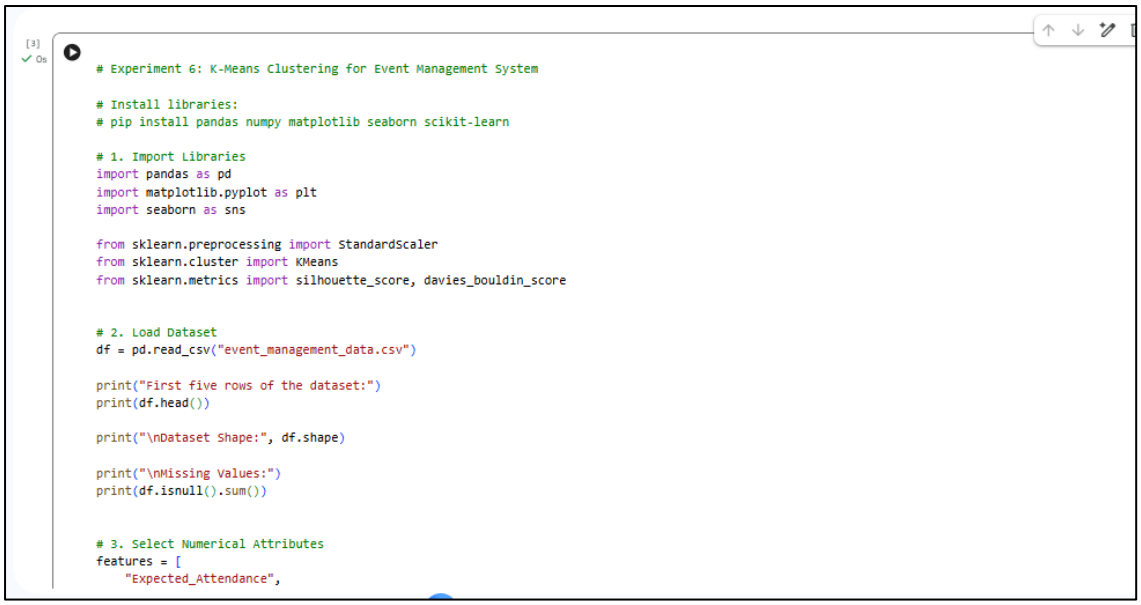
1. **Elbow Method:** Helps determine the optimal number of clusters by plotting the inertia for different values of K. The elbow point indicates a suitable number of clusters.
2. **Silhouette Score:** Measures how well each data point fits within its assigned cluster compared with other clusters. Its value ranges from -1 to 1, with higher values generally indicating better-separated clusters.

3. **Inertia:** Measures the sum of squared distances between data points and their assigned cluster centroids. Lower inertia indicates more compact clusters, although it generally decreases as K increases.
4. **Davies-Bouldin Index:** Measures the similarity between clusters using their internal distances and the distances between their centroids. A lower value generally indicates better-separated and more compact clusters.

Tools Required

1. Python 3.x
2. Jupyter Notebook or Google Colab
3. Libraries: Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn
4. Seeds dataset in CSV format

Code



```
# Experiment 6: K-Means Clustering for Event Management System

# Install libraries:
# pip install pandas numpy matplotlib seaborn scikit-learn

# 1. Import Libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score, davies_bouldin_score


# 2. Load Dataset
df = pd.read_csv("event_management_data.csv")

print("First five rows of the dataset:")
print(df.head())

print("\nDataset Shape:", df.shape)

print("\nMissing Values:")
print(df.isnull().sum())


# 3. Select Numerical Attributes
features = [
    "Expected_Attendance",
```



```
    "Budget_INR",
    "Ticket_Price_INR",
    "Duration_Hours",
    "Number_of_Activities"
]

X = df[features]


# 4. Standardize the Data
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

print("\nData standardized successfully.")


# 5. Plot the Elbow Method
inertia_values = []

for k in range(1, 11):
    model = KMeans(
        n_clusters=k,
        random_state=42,
        n_init=10
    )

    model.fit(X_scaled)
    inertia_values.append(model.inertia_)

plt.figure(figsize=(8, 5))
plt.plot(range(1, 11), inertia_values, marker="o")

plt.title("Elbow Method - Event Management")
```

```
plt.xlabel("Number of Clusters (K)")
plt.ylabel("Inertia")
plt.xticks(range(1, 11))
plt.grid(True)
plt.show()


# 6. Apply K-Means Clustering (K=3)
kmeans = KMeans(
    n_clusters=3,
    random_state=42,
    n_init=10
)

df["Cluster"] = kmeans.fit_predict(X_scaled)

print("\nEvent Cluster Assignments:")
print(df[["Event_Name", "Cluster"]].to_string(index=False))

print("\nNumber of Events in Each Cluster:")
print(df["Cluster"].value_counts().sort_index())


# 7. Calculate Evaluation Metrics
sil_score = silhouette_score(X_scaled, df["Cluster"])
inertia = kmeans.inertia_
db_score = davies_bouldin_score(X_scaled, df["Cluster"])

print("\nClustering Evaluation Metrics:")
print("Silhouette Score:", round(sil_score, 4))
print("Inertia:", round(inertia, 4))
print("Davies-Bouldin Index:", round(db_score, 4))
```

```
# 8. Visualize Clustering Results
plt.figure(figsize=(9, 6))

sns.scatterplot(
    data=df,
    x="Expected_Attendance",
    y="Budget_INR",
    hue="Cluster",
    palette="Set1",
    s=90
)

plt.title("Event Clusters: Attendance vs Budget")
plt.xlabel("Expected Attendance")
plt.ylabel("Budget (INR)")
plt.legend(title="Cluster")
plt.grid(True)
plt.tight_layout()
plt.show()


# 9. Analyze Cluster Characteristics
cluster_means = df.groupby("Cluster")[features].mean()

print("\nMean Characteristics of Each Cluster:")
print(cluster_means.round(2))


# 10. Display Events in Each Cluster
print("\nEvents Grouped by Cluster:")

for cluster_number, group in df.groupby("Cluster"):
```

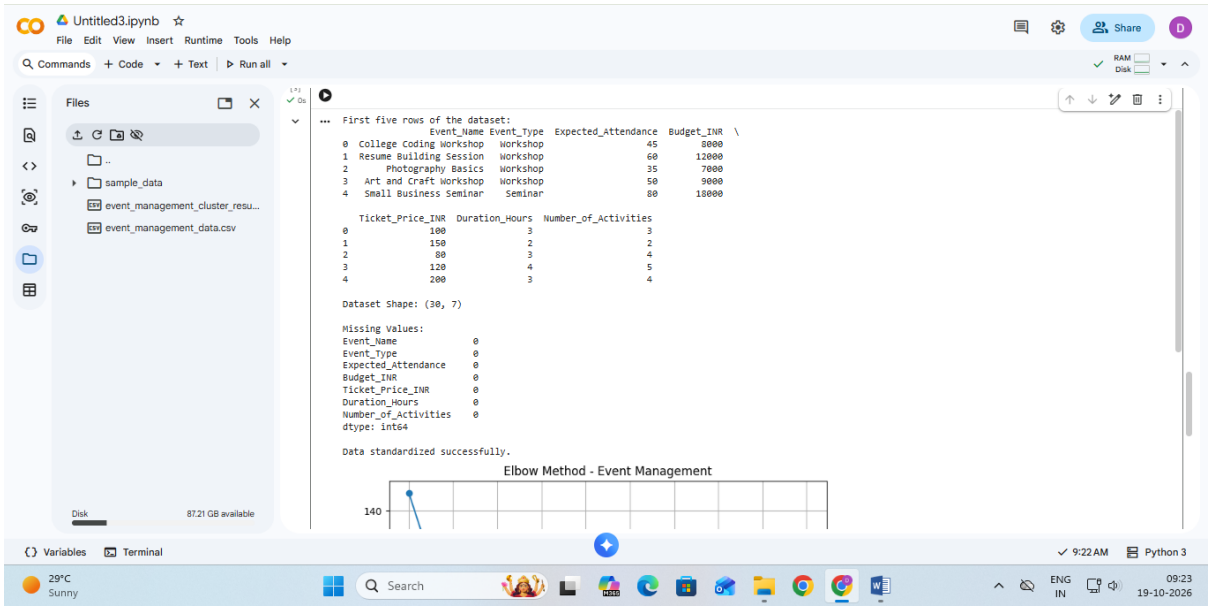
```
# 10. Display Events in Each Cluster
print("\nEvents Grouped by Cluster:")

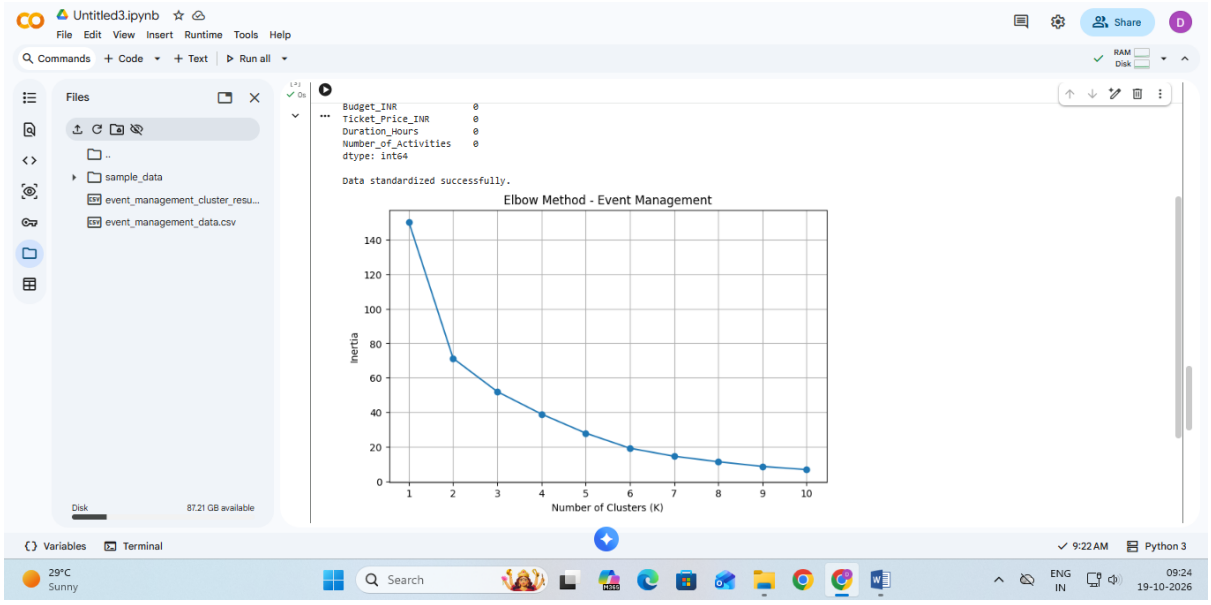
for cluster_number, group in df.groupby("Cluster"):
    print(f"\nCluster {cluster_number}:")
    print(group["Event_Name"].to_list())

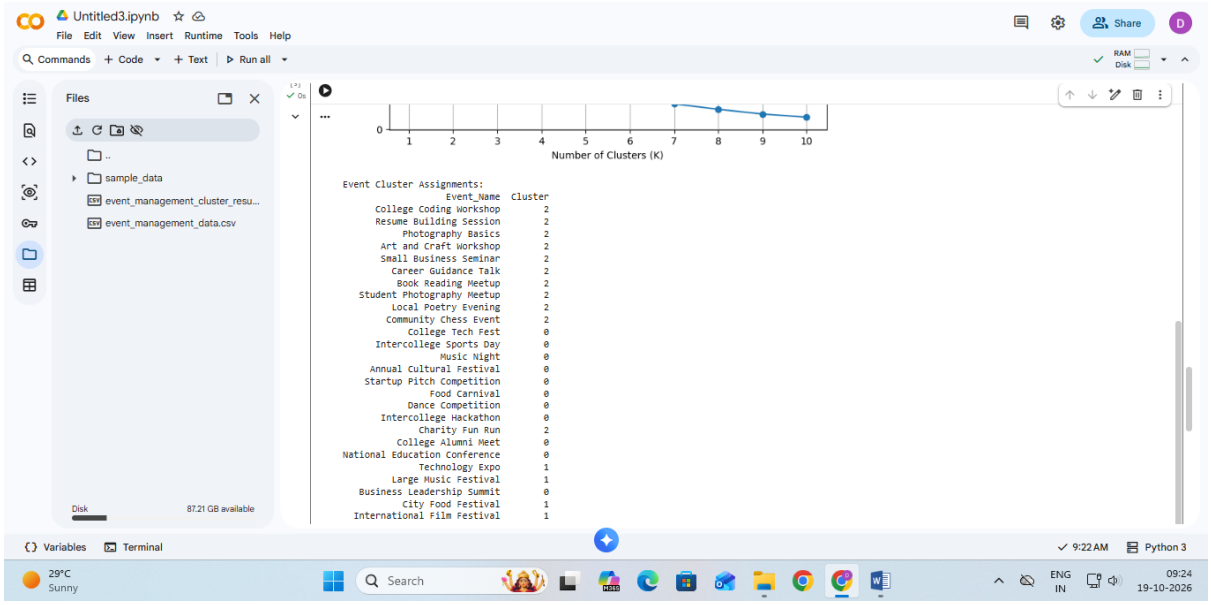

# 11. Save Results
df.to_csv("event_management_cluster_results.csv", index=False)

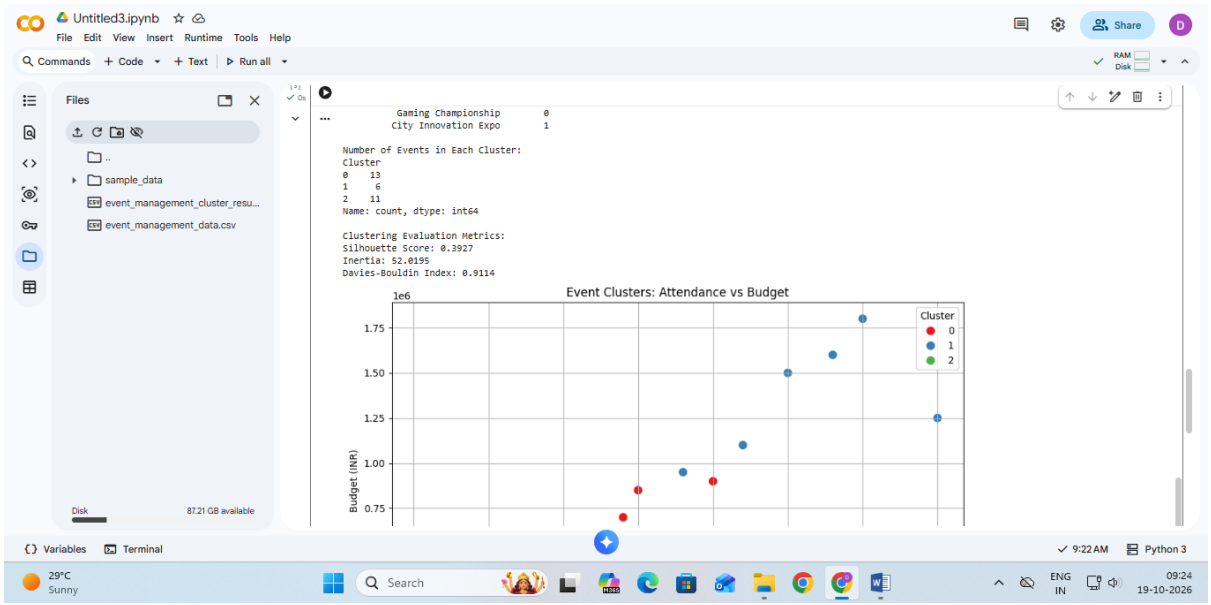
print("\nResults saved to event_management_cluster_results.csv")
print("\nExperiment Completed Successfully!")
```

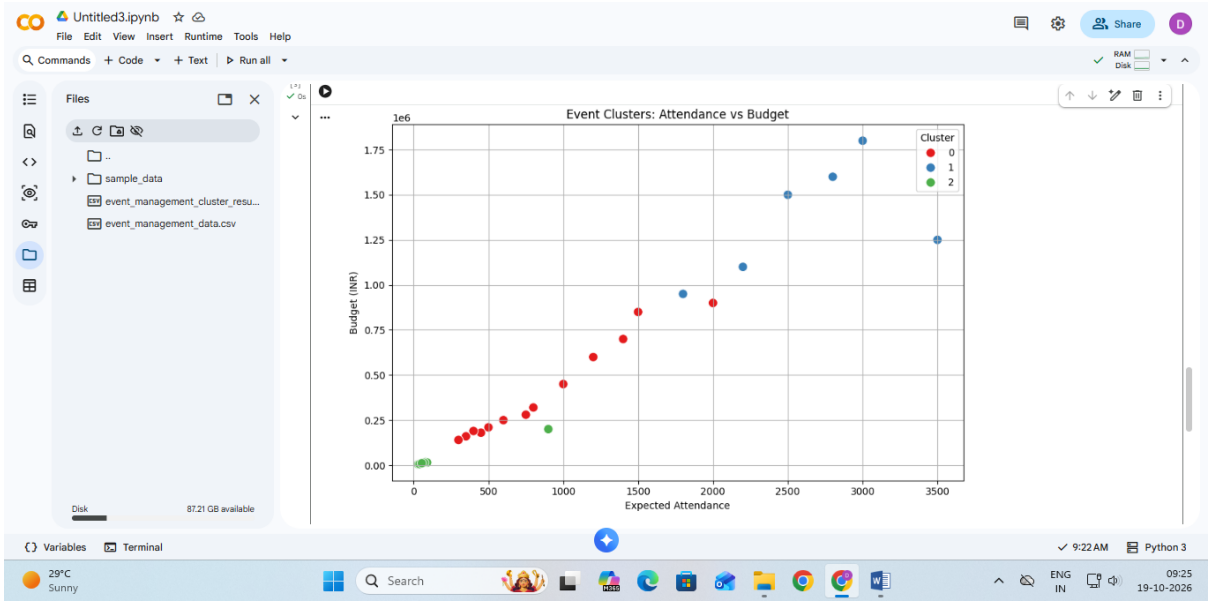
Output

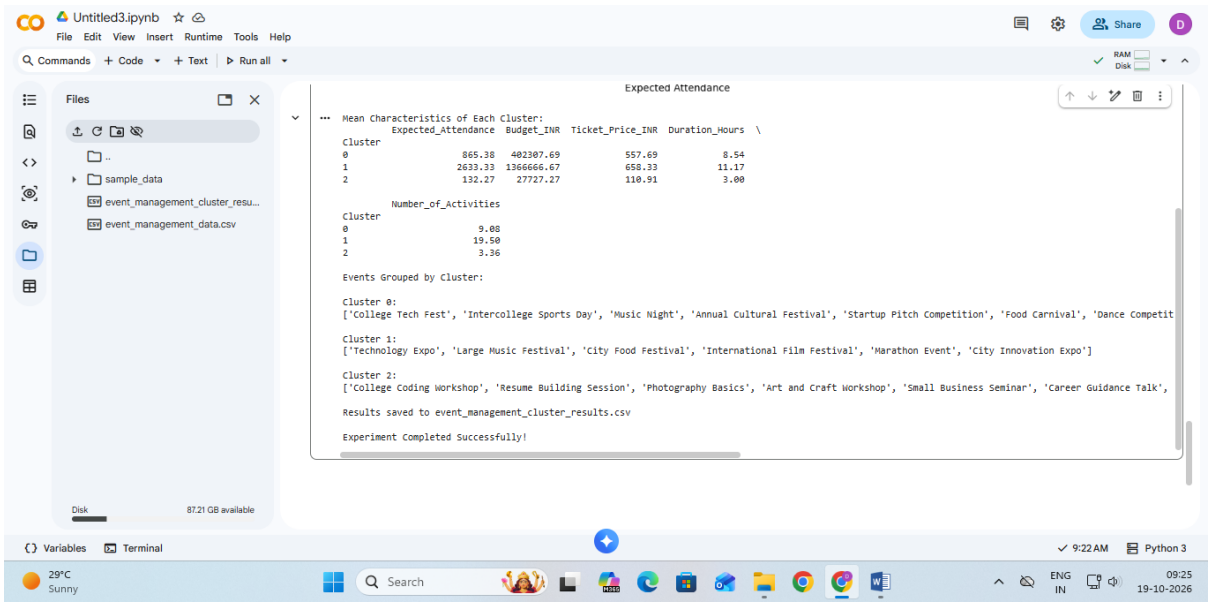

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File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
Files
sample_data
event_management_cluster_resu...
event_management_data.csv
Disk 87.21 GB available
First five rows of the dataset:
Event_Name Event_Type Expected_Attendance Budget_INR \
0 College Coding Workshop Workshop 45 8000
1 Resume Building Session Workshop 60 12000
2 Photography Basics Workshop 35 7000
3 Art and Craft Workshop Workshop 50 9000
4 Small Business Seminar Seminar 80 18000
Ticket_Price_INR Duration_Hours Number_of_Activities
0 100 3 3
1 150 2 2
2 80 3 4
3 120 4 5
4 200 3 4
Dataset Shape: (30, 7)
Missing Values:
Event_Name 0
Event_Type 0
Expected_Attendance 0
Budget_INR 0
Ticket_Price_INR 0
Duration_Hours 0
Number_of_Activities 0
dtype: int64
Data standardized successfully.
Elbow Method - Event Management
140
Variables Terminal
9:22 AM Python 3
29°C Sunny
Search
ENG IN
09:23 19-10-2026


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Files
sample_data
event_management_cluster_resu...
event_management_data.csv
Disk 87.21 GB available
Budget_INR 0
Ticket_Price_INR 0
Duration_Hours 0
Number_of_Activities 0
dtype: int64
Data standardized successfully.
Elbow Method - Event Management
Inertia
Number of Clusters (K)
0
20
40
60
80
100
120
140
1
2
3
4
5
6
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8
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10
Variables Terminal
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Commands + Code + Text Run all
Files
sample_data
event_management_cluster_resu...
event_management_data.csv
Disk 87.21 GB available
Number of Clusters (K)
Event Cluster Assignments:
Event_Name Cluster
College Coding Workshop 2
Resume Building Session 2
Photography Basics 2
Art and Craft Workshop 2
Small Business Seminar 2
Career Guidance Talk 2
Book Reading Meetup 2
Student Photography Meetup 2
Local Poetry Evening 2
Community Chess Event 2
College Tech Fest 0
Intercollege Sports Day 0
Music Night 0
Annual Cultural Festival 0
Startup Pitch Competition 0
Food Carnival 0
Dance Competition 0
Intercollege Hackathon 0
Charity Fun Run 2
College Alumni Meet 0
National Education Conference 0
Technology Expo 1
Large Music Festival 1
Business Leadership Summit 0
City Food Festival 1
International Film Festival 1
Variables Terminal
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File Edit View Insert Runtime Tools Help
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Commands + Code + Text Run all
Files
sample_data
event_management_cluster_resu...
event_management_data.csv
Disk 87.21 GB available
Gaming Championship 0
City Innovation Expo 1
Number of Events in Each Cluster:
Cluster
0 13
1 6
2 11
Name: count, dtype: int64
Clustering Evaluation Metrics:
Silhouette Score: 0.3927
Inertia: 52.0195
Davies-Bouldin Index: 0.9114
Event Clusters: Attendance vs Budget
1e6
Budget (INR)
Cluster
0
1
2
Variables Terminal
9:22 AM Python 3
29°C Sunny
Search
ENG IN
09:24 19-10-2026


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Commands + Code + Text Run all
Files
sample_data
event_management_cluster_resu...
event_management_data.csv
Disk 87.21 GB available
Event Clusters: Attendance vs Budget
1e6
Budget (INR)
Expected Attendance
Cluster
0
1
2
Variables Terminal
9:22 AM Python 3
29°C Sunny
Search
ENG IN
09:25 19-10-2026


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File Edit View Insert Runtime Tools Help
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Commands + Code + Text Run all
Files
sample_data
event_management_cluster_resu...
event_management_data.csv
Disk 87.21 GB available
Expected Attendance
Mean Characteristics of Each Cluster:
Expected_Attendance Budget_INR Ticket_Price_INR Duration_Hours \
Cluster
0 865.38 402307.69 557.69 8.54
1 2633.33 1366666.67 658.33 11.17
2 132.27 27727.27 110.91 3.00
Number_of_Activities
Cluster
0 9.08
1 19.50
2 3.36
Events Grouped by Cluster:
Cluster 0:
['College Tech Fest', 'Intercollege Sports Day', 'Music Night', 'Annual Cultural Festival', 'Startup Pitch Competition', 'Food Carnival', 'Dance Competit
Cluster 1:
['Technology Expo', 'Large Music Festival', 'City Food Festival', 'International Film Festival', 'Marathon Event', 'City Innovation Expo']
Cluster 2:
['College Coding Workshop', 'Resume Building Session', 'Photography Basics', 'Art and Craft Workshop', 'Small Business Seminar', 'Career Guidance Talk',
Results saved to event_management_cluster_results.csv
Experiment Completed Successfully!
Variables Terminal
9:22 AM Python 3
29°C Sunny
Search
ENG IN
09:25 19-10-2026

Inference

Result

Conclusion

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GitHub Repository Link:

https://github.com/Sentry2311/DMBI_EXPERIMENTS/tree/main/Experiment_06_KMeans_Clustering_Python

(Contains complete documentation, dataset, and program code for Experiment 6: K-Means Clustering Using Python)